

1. Administrative & Study Identification

Full Study Title: A Study to Learn About the Safety and Immune Response of a New Pneumococcal Vaccine in Adults **Short Title/Acronym:** Safety and Immune Response of a New Pneumococcal Vaccine in Adults **Protocol Number:** C4931001 **NCT Number:** NCT06182124 **Posting ID:** 575532955283293815 **Sponsor/Company Name:** Pfizer **Investigational Product:** Multivalent Pneumococcal Vaccine (Formulations 1, 2, 3, and 4) **Phase of Development:** Phase 1/2 (PHASE1, PHASE2) **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective

- **Primary Objective:** To evaluate the safety, tolerability, and immune effects of new multivalent pneumococcal vaccine candidates (different formulations) in adults.
- **Secondary Objectives:** To measure specific immune responses (e.g., Opsonophagocytic Activity [OPA] and Immunoglobulin G [IgG]) triggered by the vaccine candidates compared to licensed pneumococcal comparator vaccines (e.g., 20vPnC/Prevnar 20).

2.2 Background & Rationale To address the ongoing burden of pneumococcal disease by exploring next-generation multivalent pneumococcal vaccines that may provide broader serotype coverage and protection. This study evaluates multiple new vaccine formulations in healthy adults and adults with stable pre-existing conditions to determine optimal safety and immunogenicity profiles for future Phase 3 evaluation.

3. Study Design & Methodology

3.1 Design Framework

- **Study Type:** Interventional
- **Control Type:** Active Comparator (Licensed pneumococcal comparator vaccines, including 20-valent pneumococcal conjugate vaccine [20vPnC])
- **Blinding:** Double-Blind
- **Randomization:** Randomized

3.2 Planned Enrollment & Duration

- **Target \$\$\$:** Approximately 558 participants
 - **Number of Sites:** 21 locations in the United States
 - **Estimated Study Start:** 20-Dec-2023
 - **Estimated Primary Completion:** 23-May-2026
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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants ≥ 18 to < 50 years of age (Stage 1) and ≥ 50 years of age (Stage 2, Stage 3, and Stage 4) at the time of consent. 2. Adults determined by clinical assessment to be eligible for the study, including adults with preexisting stable disease within 12 weeks before receipt of study intervention. 3. Participants willing to use acceptable contraception for at least 28 days after study intervention; female participants not of childbearing potential; male participants unable to father children. 4. For Phase 1 (Stage 1): Adults 18 through 49 years of age with no history of ever receiving a pneumococcal vaccine (i.e., pneumococcal vaccine-naïve). 5. For Phase 2 (Stages 2, 3, and 4): Adults ≥ 50 through 64 years of age with no history of ever receiving a pneumococcal vaccine (i.e., pneumococcal vaccine-naïve) and adults ≥ 65 years of age and older with either no history of ever receiving a pneumococcal vaccine or who received PPSV23 >5 years prior to the first vaccination in this study (i.e., experienced).

4.2 Exclusion Criteria 1. History of severe adverse reaction associated with a vaccine and/or severe allergic reaction to any component in pneumococcal conjugate vaccines and any other diphtheria toxoid-containing vaccine. 2. Serious chronic disorder that, in the investigator's opinion, would make the participant inappropriate for entry into the study. 3. Medical or psychiatric condition (recent or active suicidal ideation/behavior or laboratory abnormality) that may increase the risk of study participation or make participant inappropriate for the study. 4. Known or suspected immunodeficiency or other conditions associated with immunosuppression. 5. Planned receipt of any licensed or investigational pneumococcal vaccine, or planned receipt of any licensed or investigational pneumococcal vaccine through study participation. 6. Receipt of any inactivated or otherwise nonlive vaccine within 14 days or any live vaccine within 28 days before administration of study intervention.

5. Intervention & Treatment Plan

5.1 Investigational Regimen

- **Dosage/Frequency:** A single injection (shot) of a randomly assigned Multivalent Pneumococcal Vaccine (Formulation 1, 2, 3, or 4) or a licensed comparator vaccine.
- **Route of Administration:** Intramuscular (IM) injection given into the upper arm muscle.
- **Duration of Treatment:** Single day of vaccination, with an observation/follow-up duration of 6 months (Stages 1, 3, and 4) or 12 months (Stage 2).

5.2 Concomitant & Prohibited Therapy

- **Permitted:** Standard allowed medications for stable preexisting conditions.
- **Prohibited:** Immunosuppressive therapies, concurrent investigational vaccines, and non-study pneumococcal vaccines during the trial.

5.3 Associated/Ancillary Pharmacological Data (*Note: Data fields registered in the trial's structured dataset record*) * **Drug Name / Preferred Name:** remifentanyl * **Trade Names:** Ultiva * **ATC Code:** N01AH06

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening/Baseline	Day 1 Informed Consent, Medical History, Randomization, Blood Draw, Single Vaccine Dose
Follow-up 1	Month 1	Safety tracking (review of 7-day post-vaccination diary), Efficacy Assessment (Blood draw for immunogenicity)	Interim Follow-up Varies 2 to 4 clinic visits and 1 phone visit for ongoing AE/SAE tracking and overall health checks
End of Study	Month 6 or 12	Final safety review, Final blood draw (if applicable), Exit Interview	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * Percentage of participants reporting prespecified local reactions (redness, swelling, and pain at the injection site) within 7 days after vaccination. * Percentage of participants reporting prespecified systemic events (fever, headache, fatigue, muscle pain, and joint pain) within 7 days after vaccination. * Percentage of participants reporting Adverse Events (AEs) within 1 month after vaccination.

7.2 Secondary & Exploratory Endpoints * Phase 1 (Stage 1) and Phase 2 (Stage 2) Only: Pneumococcal serotype-specific Opsonophagocytic Activity (OPA) Geometric Mean Titers (GMTs) at

1-month post-vaccination. * Pneumococcal serotype-specific Immunoglobulin G (IgG) Geometric Mean Concentrations (GMCs) at 1-month post-vaccination.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Patient e-Diary tracking (completed daily for 7 days post-vaccination) and HCP interviews during scheduled clinic or phone visits.
 - **Serious Adverse Events (SAEs):** Monitored continuously from the day of consent through the 6-month or 12-month follow-up period, reported within 24 hours of investigator awareness.
 - **Data Monitoring Committee (DMC):** An internal safety review committee will evaluate data continuously, particularly between the age step-down stages.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Set (all randomized participants who received at least one vaccine dose) and Evaluable Immunogenicity Population (participants with valid baseline and post-vaccination serology without major protocol deviations).
 - **Sample Size Justification:** A sample of approximately $N=558$ provides adequate power to descriptively estimate the safety profile and immunogenic non-inferiority/superiority responses against comparator vaccines.
 - **Phase Selection Criteria Type:** PHASE_REQUIREMENT
 - **Criteria Text:** Trial must be in PHASE1, PHASE2
 - **Handling of Missing Data:** Missing serological data points falling below the Lower Limit of Quantitation (LLOQ) will typically be imputed to $0.5 \times \text{LLOQ}$. Missing safety diary data are generally not imputed.
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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Centralized and onsite monitoring visits to verify informed consent procedures, source documents, e-diary adherence, and biologic sample handling. **Audit Trail:** Secure, validated electronic Data Capture (eDC) system with full audit trails and electronic Data Clarification Forms (DCFs).

11. Study Identifiers & Governance

Primary ID: C4931001 **Posting ID:** 575532955283293815 **Registry Number:** NCT06182124 **Sponsor/Lead Organization:** Pfizer **Study**

Title: A Study to Learn About the Safety and Immune Response of a New Pneumococcal Vaccine in Adults **Official Regulatory Title:** A Phase 1/2, Randomized, Double-Blind Trial of the Safety and Immunogenicity of a Multivalent Pneumococcal Conjugate Vaccine in Adults 18 Years of Age and Older **Source Level:** 5

12. Clinical Framework (Pneumococcal Specific)

Disease Area / Name: Pneumococcal Disease **Preferred Name:** Pneumococcal Infections **Semantic Type:** Disease or Syndrome **Definition:** Infections with bacteria of the species STREPTOCOCCUS PNEUMONIAE. **MeSH Code:** D011008 **SNOMED ID:** 16814004 **SNOMED Hierarchy:** 721740007, 85769006, 408637006, 371582002, 87628006, 40733004, 64572001 **Preferred UMLS Name:** Immunizing Agents **Medical Methodology:** Active Immunization / Prophylaxis **Optimization Target:** Achievement of robust, protective serotype-specific immune responses (OPA and IgG) with an acceptable reactogenicity profile.

13. Study Procedures & Methodology

Management Pathway: Standard prophylactic vaccine evaluation pathway implemented via 4 age/formulation-based stages. **Education Protocol (HCP):** Site initiation visits for vaccine handling, blinding maintenance, and adverse event grading. **Education Protocol (Patient):** Onsite training for utilizing the electronic diary to report daily temperature, local injection site reactions, and systemic symptoms. **Quality Audit Mechanism:** Real-time e-diary compliance tracking and centralized laboratory assay validation.

14. Observation & Follow-Up Schedule

Total Duration: 6 months (Stages 1, 3, and 4) or 12 months (Stage 2) **Onsite Visit Cadence:** Day 1 (Vaccination), Month 1, and End of Study (Month 6 or 12), with a total of 2 to 4 physical clinic visits. **Remote Monitoring Frequency:** At least 1 scheduled phone visit; continuous daily safety tracking for the initial post-vaccination week. **Checklist Review Interval:** e-Diary review at Month 1 follow-up visit.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				
Statistician	[Name]	_____	[DD-MMM-YYYY]				